

Trainings ▾

General Information

Lubricating Oil

ISX engines built after February 2004, (ESN first 79035023) have the lubricating oil filter head adapter (spud) staked to the lubricating oil filter housing to reduce the possibility of the adapter backing out.

The use of quality engine lubricating oils, combined with appropriate lubricating oil drain and filter change intervals, is a critical factor in maintaining engine performance and durability.

Cummins Inc. recommends the use of a high quality 15W-40 multiviscosity heavy-duty engine lubricating oil that meets the requirements of Cummins Engineering Specification CES 20071 or CES 20076 (such as Valvoline® Premium Blue® or Premium Blue® 2000). American Petroleum Institute (API) specification CH-4 can be used as an alternative to CES 20071. Oils that meet API specification CG-4 can be used, but at a reduced drain interval. Refer to Procedure 102-002 (/qs3/pubsys2/xml/en/procedures/10/10-102-002-om-ind.html) in the Operation and Maintenance Manual Industrial and Power Generation QSX15 Engines, Bulletin 3666423 or Procedure Refer to Procedure 102-002 (/qs3/pubsys2/xml/en/procedures/10/10-102-002-om-auto.html) in the Operation and Maintenance Manual Signature and ISX Engines, Bulletin 3666251. The oil grades CC, CD, CE, and CF have been obsoleted by API and **must not** be used.

Shortened drain intervals can be required with monograde oils as determined by close monitoring of the oil condition with scheduled oil sampling. Use of single-grade oils can affect engine oil control.

Synthetic engine oils, API category III, are recommended for use in Cummins® engines operating in ambient temperature conditions consistently below -25°C [-13°F]. Above this temperature, it is recommended the petroleum based multigrade lubricants be used. Synthetic 0W-30 oils that meet API category III can be used in operations where the ambient temperature **never** exceeds 0°C [32°F]. 0W-30 oils do **not** offer the same level of protection against fuel dilution as do higher multigrade oils. Higher cylinder wear can be experienced when using 0W-30 oils in high-load situations.

For further details and an explanation of engine lubricating oils for Cummins® engines, refer to Cummins Engine Oil Recommendations, Bulletin 3810340.

Additional information regarding lubricating oil availability throughout the world is available in the Engine Manufacturing Association (EMA) Lubricating Oils Data Book for Heavy-Duty Automotive and Industrial Engines. The data book can be ordered from: Engine Manufacturers Association, Two North LaSalle Street, Suite 2200, Chicago, IL, 60602; Phone: (312) 644-6610, Fax: (312) 827-8737.

The integrated Centinel™ system is a continuous lubricating oil replacement system of electromechanical design. The Centinel™ system extends lubricating oil and lubricating oil filter change intervals through continuous lubricating oil exchange. This is accomplished by injecting engine lubricating oil at a controlled rate, proportional to fuel consumed, into the fuel system for consumption. Makeup lubricating oil (fresh) is introduced into the engine at a rate equal to the lubricating oil consumed. The Centinel™ system will also monitor the lubricating oil level in the make up tank.

Lubricating Oil Filters

This engine uses a combination NanoNet® lubricating oil filter. This filter contains 2/3rd section of StrataPore® media and 1/3rd section of high efficiency NanoNet® media.

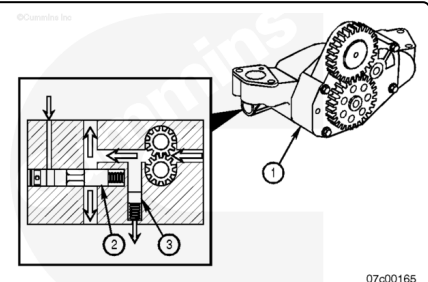
Oil flows through both StrataPore® and NanoNet® portions of the filter, with StrataPore® media providing low restriction and instantaneous protection from any damaging larger particles while the NanoNet® media prevents accumulation of the finer particles in the oil. This media combination yields a balance of protecting the engine from larger damaging particles with removal of finest particles through oil recirculation.

Lubricating oil circulates through the StrataPore® full-flow portion of the lubricating oil filter, into the lubricating oil cooler housing, and is then directed to the engine lubricating oil rifles.

A smaller portion of the lubricating oil circulates through the lower NanoNet® full flow portion of the lubricating oil filter and is directed to the engine lubricating oil rifles.

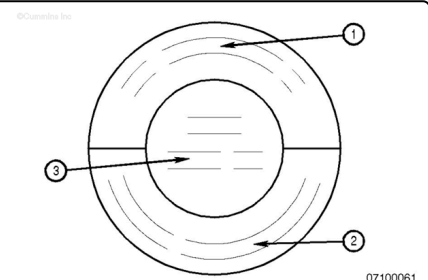
Lubricating Oil System

The demand flow and cooling lubricating system incorporates a lubricating oil pump (1) with a pressure-sensitive regulator circuit (2) and a high-pressure relief valve (3).

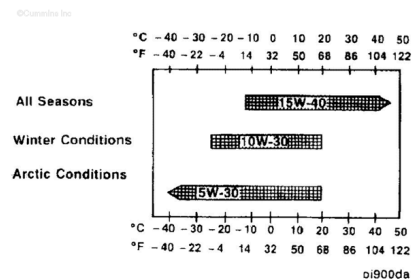


The API service symbol locations are shown in the accompanying illustration.

1. The upper half of the symbols displays the appropriate oil categories.
2. The lower half contains words to describe oil energy-conserving features.
3. The center section identifies the SAE oil viscosity grade.

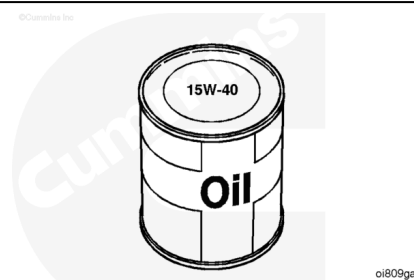


Lubricating oil viscosity **must** be chosen according to the typical climate conditions experienced by the user. Use of 15W-40 is recommended for the best engine durability at higher ambient temperature. For cold temperature conditions, 10W-30 or 5W-30 viscosity can be used for easier starting, improved lubricating oil flow, and improved fuel economy.



New Engine Break-in Oils

Special “break-in” engine lubricating oils are **not** recommended for new or rebuilt Cummins engines. In general, use the same lubricating oil during break-in as that used in normal operation. Synthetic or partially synthetic engine lubricating oils, however, can **not** be used during break-in of a new or rebuilt engine. To make sure the piston rings seat properly, use a high quality petroleum-based engine lubricating oil during the first engine lubricating oil drain period.



Oil Drain Intervals

As the engine lubricating oil becomes contaminated, essential lubricating oil additives are depleted. Lubricating oils protect the engine as long as these additives are functioning properly. Progressive contamination of the lubricating oil between lubricating oil and filter change intervals is normal. The amount of contamination will vary depending on the operation of the engine, kilometers or miles on the lubricating oil, fuel consumed, and new lubricating oil added.

Extending lubricating oil and filter change intervals beyond the recommendations will decrease engine life because of factors such as corrosion, deposits, and wear.

Use the lubricating oil drain chart in this section to determine which lubricating oil drain interval to use for an application.

